Program : Diploma in Chemical Engineering / Food Processing Technology	
Course Code : **2079**	Course Title: **Engineering Chemistry Lab**
Semester : **2**	Credits: **No Credit**
Course Category: **Engineering Science**	
Periods per week: **3 (L:3 T:0 P:0)**	Periods per semester: **45**

Course Objectives:

- To provide hands-on experience in the qualitative analysis of Inorganic compounds as simple salts and binary mixtures.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Awareness about acidic and basic radicals in Inorganic chemistry and group reagents		Applied Chemistry	1
Reaction of various acidic and basic radicals		Engineering Chemistry	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Analyse acid radicals.	9	Applying
CO2	Analyse basic radicals.	10	Applying
CO3	Analyse simple salts	12	Applying
CO4	Analyse binary mixtures.	14	Applying

CO – PO Mapping

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1		3					
CO2		3					
CO3		3					
CO4		3					3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Blooms Taxonomy Level
CO1	**Analyse acid radicals.**		
M1.01	Identify Carbonate – Sulphide – Sulphite – Sulphate – Thiosulphate – Nitrate – Fluoride – Chloride – Bromide – iodide – Acetate – Oxalate – Phosphate - Borate and Chromate.	9	Applying
CO2	**Analyse basic radicals.**		
M2.01	Identify Chromium – Cobalt – Nickel – Silver - Lead, Mercury – Copper – Cadmium – Bismuth – Arsenic – Antimony – Tin – Iron – Aluminium – Zinc – Manganese – Calcium – Barium – Strontium – Magnesium - Sodium, Potassium – Ammonium - Magnesium..	9	Applying
	Lab Exam – I	1	
CO3	**Analyse simple salts**		
M3.01	Identify the given salt, that 1. Should be soluble in water or dilute acids or conc. HCl 2. Should contain one acid radical and one basic radical 3. Should not involve the elimination of interfering acid radicals	12	Applying
CO4	**Analyse binary mixtures**		

M4.01	Identify the given binary mixture, that 1. Should be soluble in water or dilute acids or conc. HCl 2. Should contain two acids radical and two basic radicals 3. Should not involve the elimination of interfering acid radicals	9	Applying
M4.02	Open Ended Experiments	3	Applying
	Lab Exam – II	2	

Text / Reference

T/R	Book Title/Author
T1	Practical Chemistry for B.Sc Main - A.O. Thomas and Mani

Online Resources

Sl.No	Website Link
1	http://ncert.nic.in/ncerts/l/lelm107.pdf